Boolean Algebra

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Boolean Algebra

“Мы почитаем всех нулями,
А единицами — себя.”

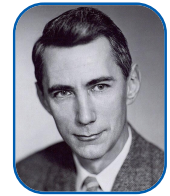
— А.С. Пушкин, «Евгений Онегин»

Gottfried Wilhelm Leibniz

George Boole

Augustus De Morgan

Charles Sanders Peirce



Claude Shannon

From Algebra to Digital Circuits

Historical note: George Boole's groundbreaking work *The Laws of Thought* (1854) created an algebraic system for logic a full century before the first electronic computers. He showed that logical reasoning could be captured by equations, replacing Aristotle's verbal arguments with mathematical symbols. This insight became the theoretical foundation for the digital age.

Key idea: Just as ordinary algebra works with numbers and operations like $+$ and $\times$, Boolean algebra works with **truth values** (0 and 1, or `false` and `true`) and operations like AND, OR, and NOT.

Historical milestones:

- **1703:** Leibniz describes binary arithmetic
- **1854:** Boole publishes algebraic logic
- **1937:** Shannon applies it to circuits
- **1940s:** First digital computers

Modern applications:

- Processor design (CPUs, GPUs)
- Formal verification (correctness proofs)
- Database optimization (queries)
- AI reasoning (SAT, planning)

Boolean Values: 0 and 1

In Boolean algebra, we work with exactly two values:

- **0** (`false`, off, low voltage)
- **1** (`true`, on, high voltage)

Example: In different contexts:

- **Mathematics:** "Is $x > 5$?" $\longrightarrow$ `true` (1) or `false` (0)
- **Programming:** `if (x && y)` $\longrightarrow$ evaluates to `true` or `false`
- **Circuits:** Wire voltage $\longrightarrow$ high (1) or low (0)
- **Sets:** Is element in set? $\longrightarrow$ yes (1) or no (0)

Why this matters: Modern computers represent all information using binary (0s and 1s).

Boolean algebra is the mathematics that makes digital computation possible.

What You Will Learn

In this lecture, you'll master Boolean algebra from foundations to applications:

1. Build Boolean expressions and evaluate them using **truth tables**
2. Prove identities using the **fundamental laws** (commutativity, De Morgan's laws, distributivity)
3. **Synthesize any Boolean function** from specifications using CNF, DNF, and Karnaugh maps
4. **Simplify complex expressions** through algebraic manipulation and graphical methods
5. Determine which operation sets are **functionally complete** using Post's criterion
6. **Design real digital circuits** with logic gates, flip-flops, and stateful elements
7. Apply Boolean algebra to **modern CS problems**: binary decision diagrams, SAT solving, and formal verification

Our goal: You'll gain both theoretical understanding (algebraic structure, formal proofs) and practical skills (circuit design, expression minimization). By the end, you'll see how the same mathematical framework powers everything from smartphone processors to AI planning algorithms.

Boolean Algebra Structure

Definition and Basic Properties

Definition 1: A *Boolean algebra* is a bounded distributive lattice $(B, \vee, \wedge, \bot, \top)$ with complement $(\cdot)' : B \to B$ such that $x \vee x' = \top$ and $x \wedge x' = \bot$.

Example: $(\mathcal{P}(A), \cup, \cap, \emptyset, A)$ with $X' = A \setminus X$ is a Boolean algebra.

Example (Digital Circuit Design): Consider 3-bit binary values as Boolean algebra:

- Elements: $\{000, 001, 010, 011, 100, 101, 110, 111\}$
- Order: Bitwise comparison ($001 \leq 011$ since $0 \leq 0$, $0 \leq 1$, $1 \leq 1$)
- Join: Bitwise OR ($010 \vee 101 = 111$)
- Meet: Bitwise AND ($110 \wedge 101 = 100$)
- Complement: Bitwise NOT ($001' = 110$)

This directly corresponds to logic gates: OR, AND, NOT gates in computer processors.

Note: Logical reading: "join" $\mapsto \vee$, "meet" $\mapsto \wedge$, "complement" $\mapsto \neg$.

Example: Database Query Lattice

Example: A database has tables `Students`, `Courses`, `Enrollments`.

- Let Q_1 = "Computer Science students"
- Let Q_2 = "Students in Math courses"
- Let Q_3 = "Graduate students"

Consider queries as lattice elements ordered by result size (specificity).

Lattice Operations:

- $Q_1 \vee Q_2$ = "Students in CS OR Math courses" (larger result set)
- $Q_1 \wedge Q_2$ = "CS students taking Math courses" (smaller result set)
- $Q_1 \wedge Q_3$ = "Graduate CS students" (most specific)

Why this matters: Query optimizers use this structure to:

1. Find equivalent but more efficient queries.
2. Cache common subqueries.
3. Predict result set sizes for cost estimation.

Complement is Unique

Theorem 1: Complements are unique in a Boolean algebra.

Proof: Suppose for some element a we have *two* complements x and y.

$$
\begin{aligned}
x &= x \wedge \top && \top \text{ is the identity for } \wedge \\
&= x \wedge (a \vee y) && \text{by definition of complement: } \top = a \vee y \\
&= (x \wedge a) \vee (x \wedge y) && \wedge \text{ distributes over } \vee \\
&= \bot \vee (x \wedge y) && \text{by definition of complement: } \ x \wedge a = \bot \\
&= (a \wedge y) \vee (x \wedge y) && \text{by definition of complement: } \ \bot = a \wedge y \\
&= (a \vee x) \wedge y && \wedge \text{ distributes over } \vee \\
&= \top \wedge y && \text{by definition of complement: } \ a \vee x = \top \\
&= y && \top \text{ is the identity for } \wedge
\end{aligned}
$$

That is, $x = y$. $\square$

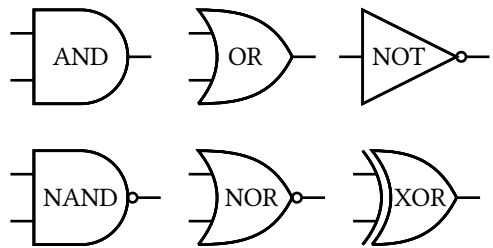
De Morgan's Laws

Theorem 2 (De Morgan): $(x \vee y)' = x' \wedge y'$ and $(x \wedge y)' = x' \vee y'$ in any Boolean algebra.

Digital Logic Circuits

Definition 2: A *logic gate* is a physical device that implements a Boolean function, taking binary inputs and producing a binary output.

Gate	Formula	Description
AND	$A \wedge B$	Outputs 1 only when both inputs are 1
OR	$A \vee B$	Outputs 1 when at least one input is 1
NOT	$\neg A$	Outputs the opposite of the input
NAND	$\neg(A \wedge B)$	Outputs 0 only when both inputs are 1
NOR	$\neg(A \vee B)$	Outputs 0 when at least one input is 1
XOR	$A \oplus B$	Outputs 1 when inputs differ
XNOR	$A \equiv B$	Outputs 1 when inputs are the same



Note: NAND and NOR gates are *universal* — any Boolean function can be implemented using only NAND gates (or only NOR gates). For example, to implement AND using NAND:

$$A \wedge B = \neg\neg(A \wedge B) = \neg(A \overline{\wedge} B) = (A \overline{\wedge} B) \overline{\wedge} (A \overline{\wedge} B)$$

Combinational Logic

Definition 3: A *combinational circuit* is a circuit where the output depends only on the current input values, without any memory or state.

Example (Half Adder): Adds two single bits:

- Sum: $S = A \oplus B$
- Carry: $C = A \wedge B$

Example (Full Adder): Adds two bits plus a carry-in:

- Sum: $S = A \oplus B \oplus C_{\text{in}}$
- Carry-out: $C_{\text{out}} = (A \wedge B) \vee (C_{\text{in}} \wedge (A \oplus B))$

Sequential Logic and Memory

Definition 4: A *sequential circuit* is a circuit where the output depends on both current inputs and previous state (memory).

Example (Flip-Flops):

- **SR Latch**: Set-Reset memory element.
- **D Flip-Flop**: Data storage triggered by clock edge.
- **JK Flip-Flop**: Eliminates forbidden state of SR latch.
- **T Flip-Flop**: Toggle flip-flop for counters.

Normal Forms

Definition 5: A *literal* is a Boolean variable or its negation (*e.g.*, x, $\neg x$).

Definition 6 (DNF): A Boolean formula is in *disjunctive normal form (DNF)* if it is a disjunction (OR) of *terms* — conjunctions (AND) of literals.

Example: $f(x, y, z) = \underbrace{(x \wedge y \wedge \neg z)}_{\text{term}} \vee \underbrace{(\neg x \wedge z)}_{\text{term}} \vee \underbrace{(\neg y \wedge \neg z)}_{\text{term}} \vee \underbrace{x}_{\text{term}}$

Definition 7 (CNF): A Boolean formula is in *conjunctive normal form (CNF)* if it is a conjunction (AND) of *clauses* — disjunctions (OR) of literals.

Example: $f(x, y, z) = \underbrace{(x \vee y \vee \neg z)}_{\text{clause}} \wedge \underbrace{(\neg x \vee z)}_{\text{clause}} \wedge \underbrace{(\neg y \vee \neg z)}_{\text{clause}} \wedge \underbrace{x}_{\text{clause}}$

Minterms and Maxterms

Definition 8 (Minterm and Maxterm):

- A *minterm* is a product (AND) of literals where each variable appears exactly once.
- A *maxterm* is a sum (OR) of literals where each variable appears exactly once.

Note: A minterm (maxterm) is a function that evaluates to 1 (0, respectively) for exactly one combination of variable values.

Example: $f(x, y, z) = x\overline{y}z$ is a minterm, and $g(x, y, z) = x + \overline{y} + z$ is a maxterm for variables x, y, z.

- $f(x, y, z) = 1$ only on input 101, *i.e.*, $x = 1$, $y = 0$, $z = 1$, corresponding to the minterm $x\overline{y}z$.
- $g(x, y, z) = 0$ only on input 010, *i.e.*, $x = 0$, $y = 1$, $z = 0$, corresponding to the maxterm $\overline{x} + y + \overline{z}$.

Canonical Forms

Definition 9 (SoP): Every Boolean function can be *uniquely* expressed as a *sum of minterms* (SoP, Sum of Products) corresponding to rows where the function evaluates to 1.

Definition 10 (PoS): Every Boolean function can be *uniquely* expressed as a *product of maxterms* (PoS, Product of Sums) corresponding to rows where the function evaluates to 0.

Karnaugh Maps

Definition 11: A *Karnaugh map* (K-map) is a graphical method for simplifying Boolean expressions by visually identifying adjacent minterms that can be combined.

AB \ CD	**00**	**01**	**11**	**10**
00	**0**	**1**	**3**	**2**
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

Zhegalkin Polynomials

Definition 12: A *Zhegalkin polynomial* is a representation of a Boolean function as a polynomial over $\mathrm{GF}(2)$ using XOR ($\oplus$) and AND ($\wedge$, often omitted) operations.

Theorem 3: Every Boolean function has a unique representation as a Zhegalkin polynomial:

$$f(x_1, \ldots, x_n) = \bigoplus_{S \subseteq \{1,\ldots,n\}} \left(a_S \prod_{i \in S} x_i \right)$$

where $a_S \in \{0, 1\}$ and $\oplus$ denotes XOR.

Example: $f(x, y) = x \vee y = x \oplus y \oplus xy$

Binary Decision Diagrams (BDDs)

Definition 13 (BDD): A *binary decision diagram (BDD)* is a directed acyclic graph representing a Boolean function, where each non-terminal node represents a variable test and edges represent variable assignments.

Definition 14 (ROBDD): A *reduced* ordered binary decision diagram (ROBDD) is an ordered BDD with a fixed variable ordering where:

- No variable appears more than once on any path
- No two nodes have identical low and high successors
- No node has identical low and high successors

Theorem 4: Every Boolean function has a unique reduced ordered binary decision diagram (ROBDD) representation for a given variable ordering.

TODO

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